

## CAPE SAGRO, France

WMO: 077910

Lat: 42.798N

Lon: 9.489E

Elev: 120

StdP: 99.89

Time Zone: 1.00 (EUC)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-----|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |     |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p) |
| 2             | 3.8        | 5.0 | -6.1                          | 2.3 | 6.9  | -4.0 | 2.7 | 7.5  | 27.4                  | 15.6 | 24.9 | 13.5 | 4.5                   | 300  | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 4.7                    | 29.2            | 22.7 | 28.4 | 22.6 | 27.6 | 22.3 | 24.8                | 27.7 | 24.2 | 27.0 | 23.7 | 26.4 | 3.1                  | 20   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.9                            | 19.0 | 26.9 | 23.2 | 18.3 | 26.4 | 22.7 | 17.6 | 25.9 | 76.1          | 27.8 | 73.6 | 27.2 | 71.4 | 26.4 | 27.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 21.1          | 15.3 | 11.7 | DB                         | 1.5  | 30.9               | 2.1 | 0.8                                                | 0.0  | 31.5       | -1.2 | 32.0       | -2.4 | 32.5       | -3.9 | 33.1 | (4) |
| (5)               |      |      | WB                         | -0.3 | 25.6               | 1.5 | 1.0                                                | -1.4 | 26.3       | -2.2 | 26.9       | -3.1 | 27.4       | -4.2 | 28.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 16.8        | 10.4  | 10.2  | 11.6  | 14.3  | 18.0  | 20.8  | 24.8  | 24.9  | 22.1  | 18.8  | 14.3  | 11.4  | (6)  |      |
|                                                                        | DBStd       | 5.73        | 2.35  | 2.69  | 2.64  | 1.81  | 2.47  | 2.13  | 1.66  | 1.46  | 2.03  | 2.15  | 2.57  | 2.66  | (7)  |      |
|                                                                        | HDD10.0     | 88          | 24    | 28    | 17    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 16    | (8)  |      |
|                                                                        | HDD18.3     | 1203        | 246   | 228   | 209   | 122   | 35    | 4     | 0     | 0     | 2     | 20    | 123   | 214   | (9)  |      |
|                                                                        | CDD10.0     | 2579        | 36    | 33    | 66    | 128   | 247   | 323   | 460   | 463   | 362   | 271   | 130   | 60    | (10) |      |
|                                                                        | CDD18.3     | 656         | 0     | 0     | 0     | 1     | 24    | 77    | 202   | 204   | 114   | 33    | 1     | 0     | (11) |      |
|                                                                        | CDH23.3     | 3154        | 0     | 0     | 0     | 0     | 39    | 163   | 1344  | 1299  | 301   | 8     | 0     | 0     | (12) |      |
| (13)                                                                   | CDH26.7     | 393         | 0     | 0     | 0     | 0     | 3     | 7     | 202   | 169   | 12    | 0     | 0     | 0     | (13) |      |
| (14)                                                                   | Wind        | WSAvg       | 3.8   | 3.9   | 4.0   | 4.1   | 3.9   | 3.1   | 3.4   | 3.0   | 3.1   | 3.3   | 4.4   | 4.8   | 4.7  | (14) |
| (15) Precipitation                                                     | PrecAvg     | 622         | 53    | 49    | 55    | 56    | 38    | 28    | 10    | 17    | 54    | 85    | 110   | 68    | (15) |      |
|                                                                        | PrecMax     | 873         | 159   | 149   | 144   | 116   | 126   | 94    | 57    | 66    | 155   | 148   | 230   | 137   | (16) |      |
|                                                                        | PrecMin     | 429         | 7     | 0     | 4     | 7     | 5     | 2     | 0     | 0     | 1     | 10    | 19    | 7     | (17) |      |
|                                                                        | PrecStd     | 123         | 32    | 35    | 35    | 29    | 32    | 23    | 14    | 16    | 45    | 43    | 58    | 39    | (18) |      |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 17.0  | 17.7  | 18.5  | 20.9  | 26.4  | 27.8  | 30.7  | 30.3  | 27.9  | 24.1  | 20.7  | 19.7  | (19) |      |
|                                                                        |             | MCWB        | 11.7  | 10.9  | 11.5  | 15.7  | 19.4  | 21.9  | 21.5  | 22.8  | 21.8  | 18.8  | 16.8  | 13.5  | (20) |      |
|                                                                        | 2%          | DB          | 15.1  | 16.1  | 16.8  | 19.4  | 24.1  | 26.4  | 29.6  | 29.1  | 26.7  | 23.1  | 19.2  | 17.0  | (21) |      |
|                                                                        |             | MCWB        | 11.1  | 10.8  | 11.8  | 15.3  | 19.2  | 21.5  | 22.5  | 22.8  | 21.1  | 18.8  | 16.1  | 13.0  | (22) |      |
|                                                                        | 5%          | DB          | 14.0  | 14.7  | 15.7  | 18.2  | 22.7  | 25.2  | 28.7  | 28.3  | 25.9  | 22.2  | 18.2  | 15.6  | (23) |      |
|                                                                        |             | MCWB        | 10.9  | 10.8  | 11.7  | 14.0  | 18.2  | 21.1  | 22.7  | 22.7  | 21.0  | 18.6  | 15.0  | 12.5  | (24) |      |
|                                                                        | 10%         | DB          | 13.3  | 13.5  | 14.8  | 17.1  | 21.6  | 24.1  | 27.8  | 27.7  | 25.1  | 21.3  | 17.3  | 14.8  | (25) |      |
|                                                                        |             | MCWB        | 10.4  | 10.2  | 11.4  | 13.6  | 17.4  | 20.5  | 22.5  | 22.4  | 20.5  | 18.1  | 14.3  | 11.8  | (26) |      |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 13.4  | 13.4  | 14.1  | 18.0  | 21.7  | 24.2  | 25.7  | 25.7  | 24.4  | 21.4  | 18.4  | 15.0  | (27) |      |
|                                                                        |             | MCDB        | 15.1  | 15.4  | 15.4  | 19.9  | 23.7  | 26.5  | 28.5  | 28.6  | 26.3  | 22.8  | 19.8  | 16.9  | (28) |      |
|                                                                        | 2%          | WB          | 12.4  | 12.2  | 13.3  | 16.1  | 20.4  | 23.1  | 25.0  | 24.8  | 23.2  | 20.5  | 17.0  | 14.1  | (29) |      |
|                                                                        |             | MCDB        | 14.0  | 14.3  | 14.7  | 18.5  | 23.2  | 25.2  | 27.8  | 27.8  | 25.3  | 21.9  | 18.8  | 16.0  | (30) |      |
|                                                                        | 5%          | WB          | 11.8  | 11.6  | 12.9  | 15.0  | 19.2  | 22.1  | 24.4  | 24.2  | 22.4  | 19.8  | 15.9  | 13.3  | (31) |      |
|                                                                        |             | MCDB        | 13.5  | 13.6  | 14.3  | 17.1  | 22.1  | 24.5  | 27.1  | 27.2  | 24.6  | 21.3  | 17.6  | 15.2  | (32) |      |
|                                                                        | 10%         | WB          | 11.0  | 10.9  | 12.4  | 14.3  | 18.3  | 21.2  | 23.8  | 23.7  | 21.6  | 19.1  | 15.0  | 12.2  | (33) |      |
|                                                                        |             | MCDB        | 12.9  | 13.0  | 14.1  | 16.4  | 20.7  | 23.6  | 26.5  | 26.6  | 23.9  | 20.8  | 16.9  | 14.1  | (34) |      |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 3.6   | 3.9   | 3.9   | 4.0   | 4.4   | 4.5   | 4.7   | 4.7   | 4.6   | 3.9   | 3.5   | 3.5   | (35) |      |
|                                                                        |             | MCDBR       | 4.3   | 4.8   | 5.0   | 5.4   | 5.8   | 5.2   | 5.4   | 5.3   | 5.1   | 4.5   | 3.9   | 4.6   | (36) |      |
|                                                                        | 5% WB       | MCWBR       | 3.4   | 3.9   | 4.0   | 3.9   | 4.3   | 3.9   | 4.1   | 4.1   | 3.8   | 3.5   | 3.5   | 3.7   | (37) |      |
|                                                                        |             | MCWBR       | 3.7   | 4.0   | 3.7   | 4.4   | 4.9   | 4.6   | 4.7   | 4.8   | 4.7   | 3.5   | 3.2   | 4.5   | (38) |      |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.315 | 0.330 | 0.356 | 0.394 | 0.403 | 0.401 | 0.398 | 0.396 | 0.385 | 0.364 | 0.342 | 0.316 | (40) |      |
|                                                                        |             | taud        | 2.547 | 2.503 | 2.446 | 2.345 | 2.364 | 2.407 | 2.412 | 2.432 | 2.450 | 2.510 | 2.521 | 2.550 | (41) |      |
|                                                                        | Edn at Noon | Edn at Noon | 832   | 874   | 887   | 874   | 872   | 872   | 870   | 862   | 846   | 822   | 787   | 795   | (42) |      |
|                                                                        |             | Edh at Noon | 69    | 84    | 100   | 119   | 120   | 116   | 114   | 108   | 99    | 83    | 71    | 64    | (43) |      |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.68  | 2.56  | 3.81  | 5.14  | 6.23  | 7.04  | 7.17  | 6.25  | 4.70  | 3.07  | 1.88  | 1.48  | (44) |      |
|                                                                        | RadStd      | RadStd      | 0.15  | 0.26  | 0.41  | 0.31  | 0.46  | 0.35  | 0.30  | 0.29  | 0.21  | 0.27  | 0.17  | 0.17  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (19 neighbors) | +0.22 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +61     | N/A     |

Nomenclature: See separate page